

ABDOMINAL HERNIA

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Aetiology

Definition:

- A hernia is an abnormal protrusion of a cavity's contents, through a weakness in the wall of the cavity, taking with it all the linings of the cavity.

Types: Hernias may exploit natural openings :

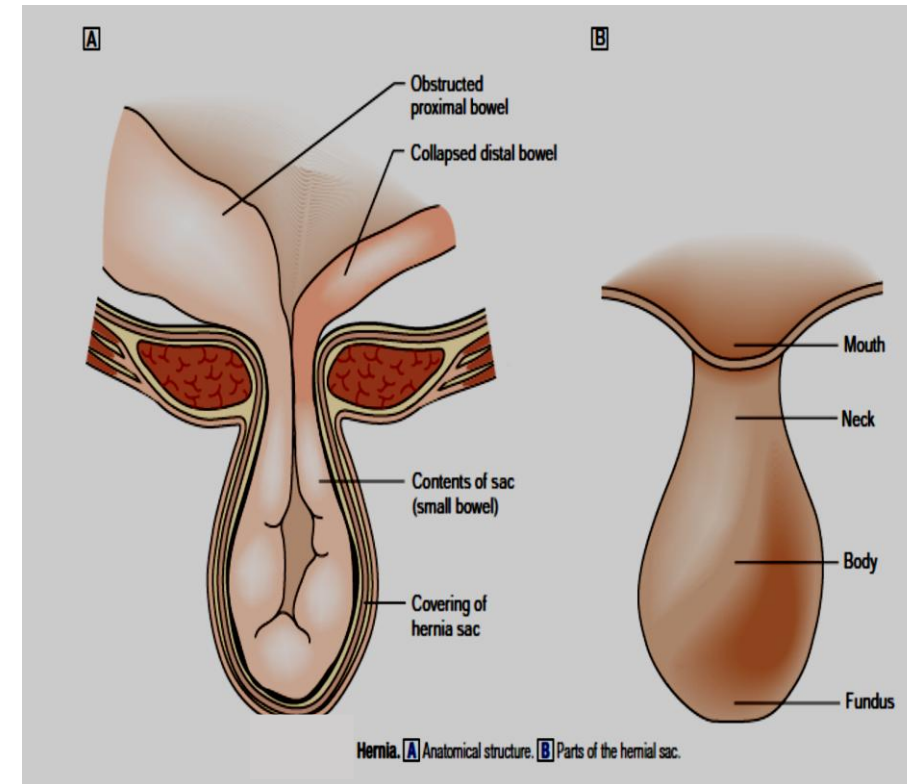
- Inguinal and femoral canals, obturator canal
- Umbilicus, esophageal hiatus, or protrude through areas weakened by stretching (e.g. epigastric hernia) or surgical incision.

Factors contribute to the development of hernias:

hernias can be considered design faults, either **anatomical** or through **inherited collagen disorders**. The **Type I to III collagen** ratio is important to make up of the tissues.

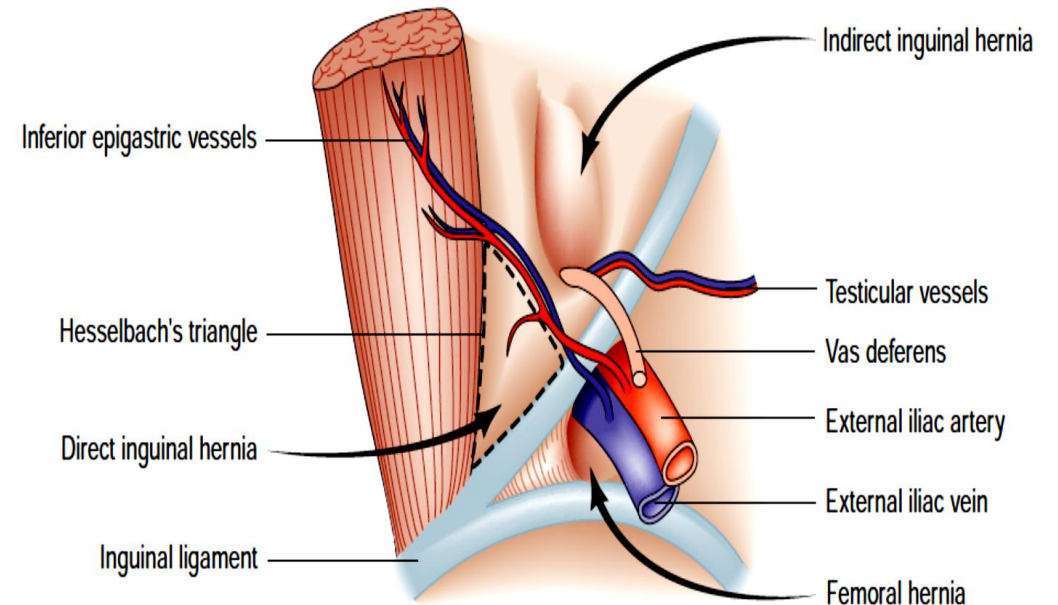
General Considerations

- **The neck:** The neck of the sac is the constriction formed by the orifice in the abdominal wall through which the hernia passes.
- **Content:** A hernia may contain most commonly omentum and/or small bowel.
- **Classification:** Hernia may be reducible or irreducible, and the contents (e.g. bowel) may become obstructed or strangulated
- **Strangulation:** denotes compromise of the blood supply of the contents and its development significantly increases morbidity and mortality.



Inguinal Hernia

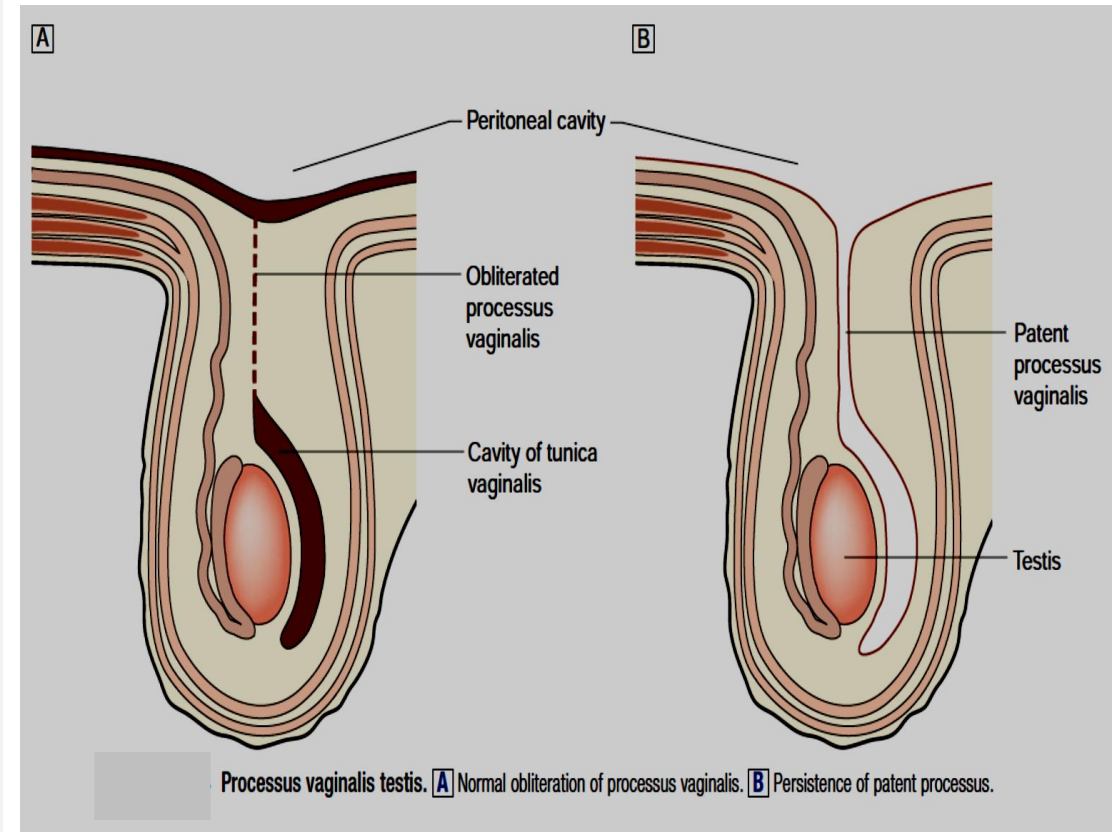
- **Incidence:** The most common types of groin hernia are indirect inguinal (60%), direct inguinal (25%) and femoral (15%).
- **Gender:** Most (85%) groin hernias occur in males.
- **Age:** In early life, an indirect inguinal hernia is the most common variety.
-After middle age, weakness of the abdominal musculature leads to an increasing incidence of **direct inguinal hernias**.
- **Femoral hernias:** are more common in females (because of stretching of widening of the femoral ring in pregnancy), but an indirect inguinal hernia is still the most common type of groin hernia in women.



Anatomy of the internal inguinal ring, showing sites of herniation from within.

Surgical anatomy

- **The inguinal canal** is an oblique passage in the lower anterior abdominal wall, through which the spermatic cord passes to the testis in the male, or the round ligament to the labium majus in the female.
- **The processus vaginalis** traversing the canal is normally obliterated at birth, but persistence in whole or in part presents an anatomical predisposition to an indirect inguinal hernia.
- The openings of the canal are formed by the internal and external rings.
- **The internal (deep) inguinal ring** lies approximately 1 cm above the mid-inguinal point (midway between the pubic tubercle and the anterior superior iliac spine).
- The internal inguinal ring is bounded medially by the inferior epigastric artery.
- **The external (superficial) inguinal ring:** The inguinal canal ends at which is an opening just above and medial to the pubic tubercle.



Coverings and Contents of Spermatic Cord

- **Coverings of the Spermatic Cord:**

- 1-The innermost layer is the internal spermatic fascia,
- 2- The middle layer is the cremasteric muscle and fascia
- 3- The outer layer is the external spermatic fascia.

- **Contents of the spermatic cord:**

- 1-The vas deferens

- 2-Arteries

- Artery of the vas , Testicular artery, Cremasteric artery

- 3- Veins:

- Pampiniform plexus of veins

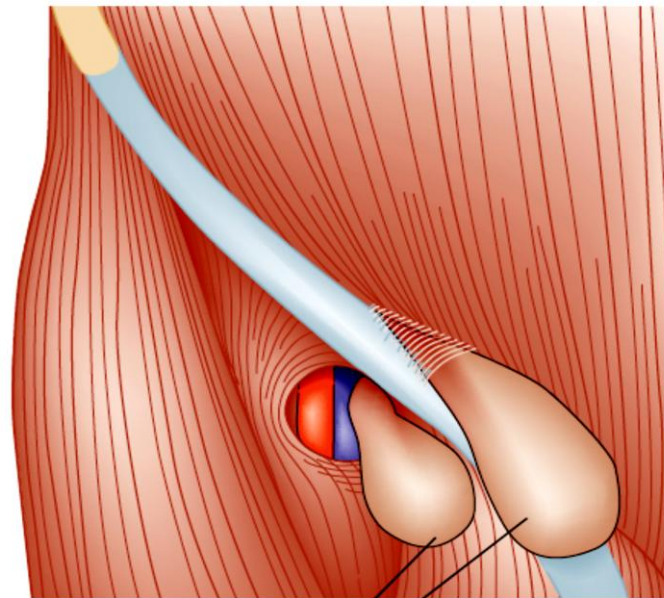
- 4-Nerves

- Ilio-inguinal nerve, the genital branch of the genitofemoral nerve

- 5-Lymphatics.

Indirect inguinal hernia

- An indirect inguinal hernia enters the internal (deep) inguinal ring and descends within the coverings of the spermatic cord so that it can pass on down into the scrotum, the so called inguino-scrotal hernia.



Femoral hernia
Indirect inguinal hernia

- **An inguinal hernia**, which passes into the scrotum, passes **above** and medial to the pubic tubercle, in contrast to **a femoral hernia**, which bulges **below** and lateral to the tubercle.



- While bilateral inguinal hernias are not common, it is unusual for both hernias to be of similar size.

Clinical Features

-
- There may be a **dragging discomfort in the groin**.
 - The hernia forms a **lump in the groin** (a swelling in the inguinal canal), which may extend into the scrotum.
 - It is often readily visible when the patient stands or is asked to cough.
 - **A cough impulse** is normally palpable, and bowel sounds can often be heard within the hernia on auscultation.
 - The hernia **often reduces spontaneously** when the patient lies down, or it may be reduced by gentle pressure applied in an upward and lateral direction.
 - It may be possible to control the hernia, once reduced, by placing a finger over the internal (deep) inguinal ring.

Direct inguinal hernia

- **Causes:** Direct hernias are due to weakness of the abdominal wall and may be precipitated by increases in intra-abdominal pressure (e.g. obstructive airways disease, BPH, chronic constipation).
- **Boundaries: (Hesselbach's triangle)**
 - *Above by the conjoint tendon*
 - *Below by the inguinal ligament*
 - *Laterally by the inferior epigastric vessels.*
- ***The sac has a wide neck, so that the hernia seldom becomes irreducible, obstructs or strangulates.***
- The neck of the sac of a direct inguinal hernia lies **medial** to the inferior epigastric vessels, whereas that of an indirect hernia lies **lateral** to them.

Management of uncomplicated inguinal hernia

- The identification of an inguinal hernia in **any child** is *nearly always* an indication to operate.
- Elective surgery is usually undertaken on a day-case basis, with liberal use of local anaesthetic blocks for postoperative pain relief.

Laparoscopic inguinal hernia repair

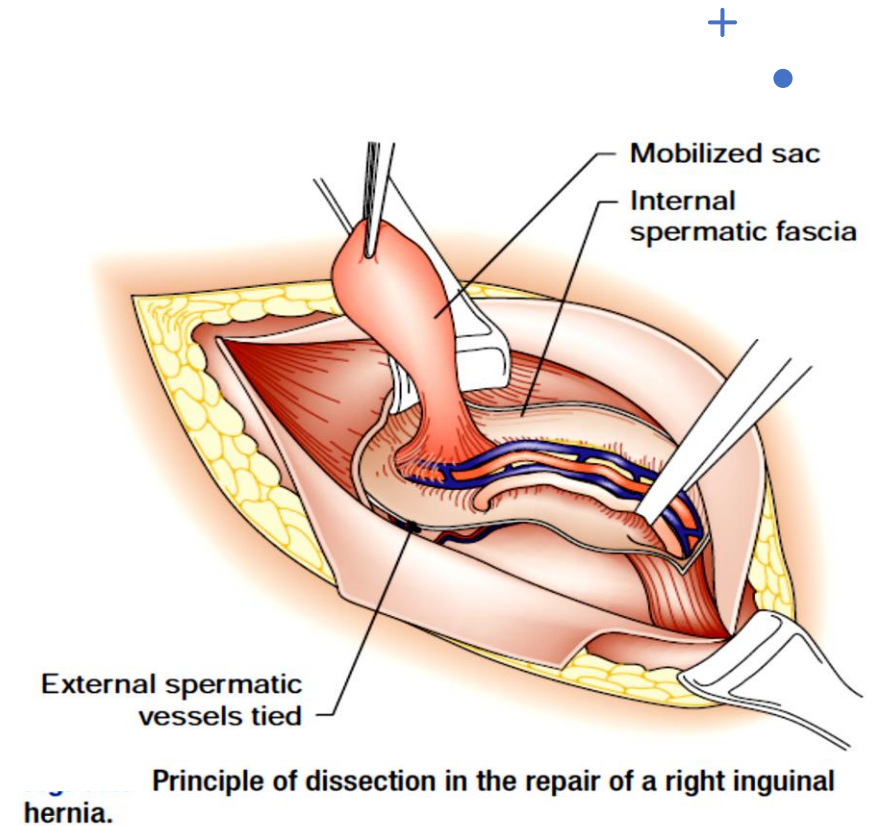
'Laparoscopic inguinal hernia repair is associated with less acute and chronic pain, less nerve injuries such as numbness, quicker return to normal activity and work, and significantly fewer postoperative complications such as infection and haematoma formation than open inguinal hernia repair. Hospital costs are higher for laparoscopic repair and it requires the use of general anaesthetic.'

Use of mesh in hernia repair

'Mesh reduces the risk of hernia recurrence.'

Indirect inguinal hernia repair

- The first step in the open approach is to open the inguinal canal, free the hernial sac from the spermatic cord and excise it after transfixing and ligating its neck.
- **Herniotomy:** Simple excision of the sac is all that is needed in young children.
- **Herniorrhaphy or Hernioplasty:** In older children and adults, the internal ring is usually stretched and widened, and therefore after herniotomy, it is necessary to tighten the deep ring and/or strengthen the posterior wall with a synthetic mesh
- Suture repair alone is rare in developed countries in adults, but still has a place in the repair of groin hernias in adolescents.



Comparison between indirect and direct inguinal hernias

	Indirect	Direct
Age	Any age, but usually young	Uncommon in children and young adult
Aetiology	Congenital (patent processus vaginalis)	Acquired weakness in abdominal wall
Relationship to inferior epigastric artery	Lateral	Medial
Descending to scrotum	Often	Rarely
Occluding the internal ring	Controls it	Does not control it
Neck	Narrow	Wide
Strangulation	More likely	Rare
Treatment	Infant—herniotomy (ligation and excision of the sac) Adult—open mesh repair, laparoscopic repair	Open mesh repair, laparoscopic repair

Femoral hernia

Key facts

- Commoner in women than men.
- Occurs through tissues of femoral canal.
- Has a high risk of strangulation due to the neck of the sac having bony and ligamentous structures limiting it on three sides.
- Approximately 30% of femoral hernias present as emergencies; 50% of these require bowel resection for strangulation and ischaemia.

Clinical features

- Appears below and lateral to pubic tubercle, medial to femoral pulse.
- May be asymptomatic until incarceration or strangulation occurs.
- May be mistaken for an upper medial thigh swelling.

Key revision points—anatomy of the femoral canal

- Lies medial to femoral vein within femoral sheath.
- Contains loose areolar tissue and a lymph node known as the lymph node of Cloquet.
- The femoral ring is the abdominal opening of the femoral canal. The increased diameter of the true pelvis in females proportionally widens the femoral canal.
- Boundaries to the femoral ring are:
 - Anteriorly, inguinal ligament.
 - Medially, lacunar ligament.
 - Posteriorly, pectineal ligament.
 - Laterally, femoral vein.

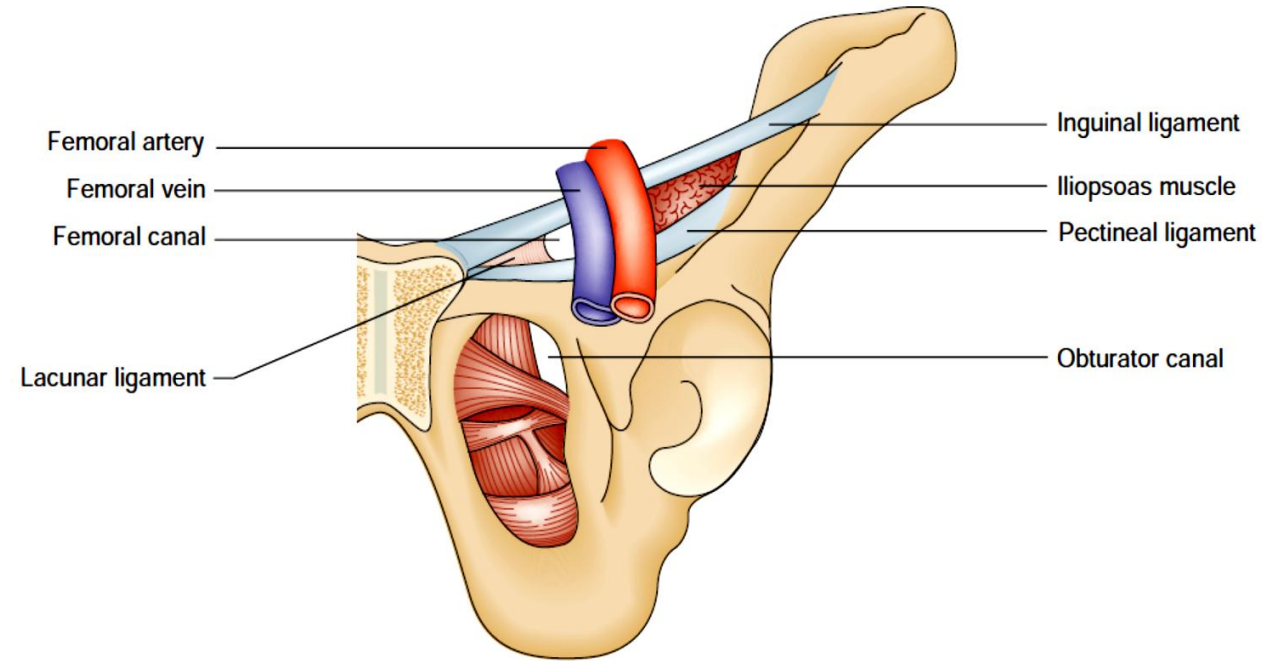


Fig. 11.9 Anatomy of the femoral ring.

Diagnosis and investigations

- Differential diagnosis includes:
 - Low presentation of inguinal hernia.
 - Femoral canal lipoma.
 - Femoral lymph node.
 - Saphena varix (compressible, disappears when lying flat, palpable thrill on coughing or percussion of the saphenous vein).
 - Femoral artery aneurysm (pulsatile).
 - Psoas abscess (fluctuant and lateral to femoral artery).
- Ultrasound scanning may help with the differential diagnosis. If there is significant doubt, exploration is usually indicated due to the high risk of complications in untreated femoral hernia.

Treatment

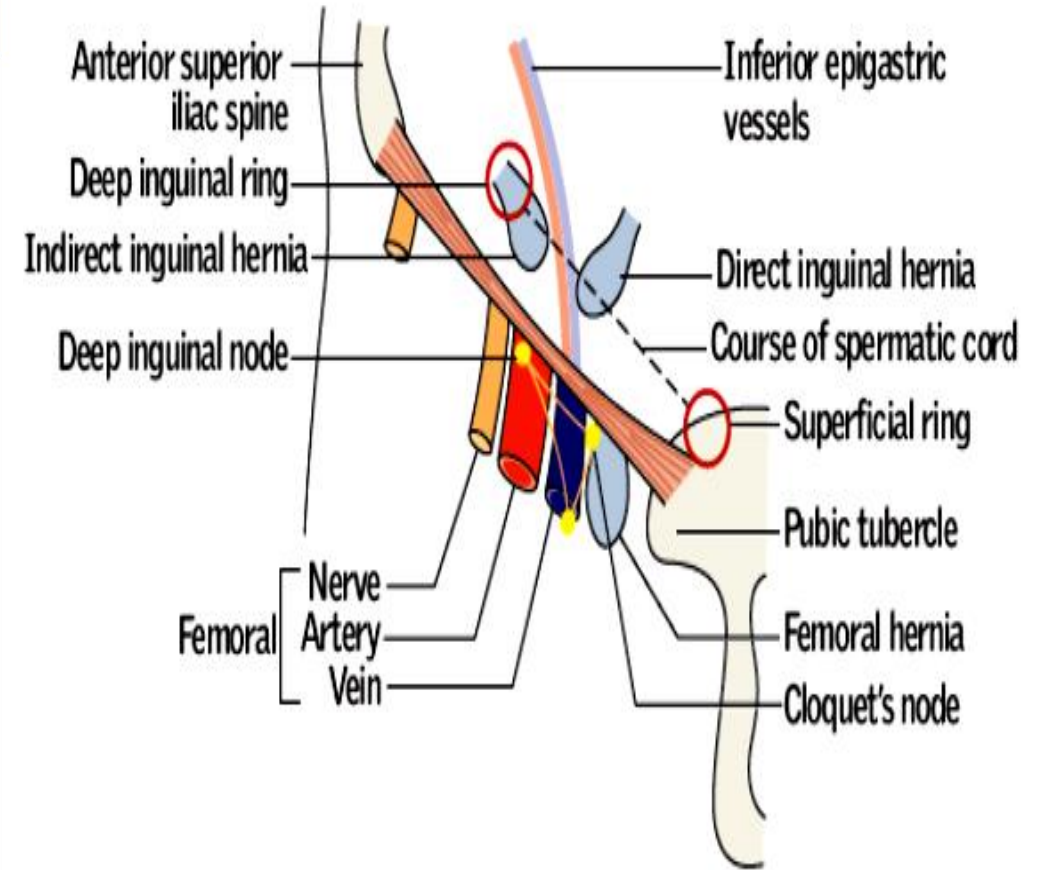
- All should be repaired because of great risk of strangulation; truss has no place in the management.
- Once the hernia is reduced, the femoral canal should be narrowed by interrupted sutures to prevent recurrence; care must be taken not to narrow the adjacent femoral vein.
- Femoral hernia repair (open) involves suture-plication of the inguinal and pectineal ligaments or placement of a mesh plug that is fixed in position in the defect with non-absorbable sutures.



SUMMARY

Groin hernias

- Indirect inguinal hernias comprise 60% of all groin hernias and commence at the deep inguinal ring, lateral to the inferior epigastric vessels
- Direct inguinal hernias account for 25% of all groin hernias and bulge through a weakness in the back wall of the inguinal canal, medial to the inferior epigastric vessels. They rarely obstruct or strangulate
- Indirect inguinal hernias may pass down within the coverings of the spermatic cord to the scrotum; direct hernias do not descend into the scrotum
- Asymptomatic inguinal hernias do not have to be repaired, especially in the elderly
- Femoral hernias account for 15% of all groin hernias and pass through the femoral canal, emerging below and lateral to the pubic tubercle (in contrast to inguinal hernias, which pass medially to the tubercle and may descend to the scrotum)
- Femoral hernias are often small and easy to miss on clinical examination, but are prone to obstruct and strangulate.



Anatomy of the inguinal canal. **Floor**: Inguinal ligament and lacunar ligament medially; **roof**: Fibres of transversalis, internal oblique; **Anterior**: External oblique aponeurosis + internal oblique for the lateral 1/3; **Posterior**: Laterally, transversalis fascia; medially, conjoint tendon.

Ventral Hernias

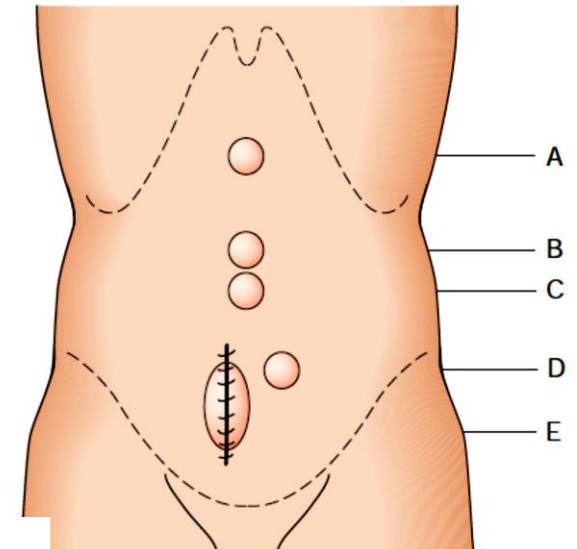
Ventral hernias occur through areas of weakness in the anterior abdominal wall: namely, the linea alba (epigastric hernia), the umbilicus (umbilical and paraumbilical hernia), the lateral border of the rectus sheath (Spigelian hernia), and the scar tissue of surgical incisions (incisional hernia). Such incisions include scars from laparoscopic surgery, the so-called port-site hernia.

Epigastric hernia

Epigastric hernias protrude through the linea alba above the level of the umbilicus.

The herniation may consist of extraperitoneal fat or may be a protrusion of peritoneum containing omentum.

The hernia is common in thin individuals.



Types of ventral hernia. [A] Epigastric: through the linea alba. [B] Umbilical: through an umbilical scar. [C] Para-umbilical: above or below the umbilicus. [D] Spigelian: lateral edge of rectus muscle. [E] Incisional: anywhere.

Para-umbilical hernia

- This hernia is caused by gradual weakening of the tissues around the umbilicus.
- It most often affects obese multiparous women, and passes through the attenuated linea alba just above or below the umbilicus.
- The hernia gradually enlarges, the covering tissues become stretched and thin, and eventually loops of bowel may become visible under parchment-like skin.
- The sac is often multilocular and may be irreducible because of adhesions that form between omentum and loops of bowel.



Clinical photograph of a para-umbilical hernia – note the appendectomy scar.

Incisional hernia

- **Over half of incisional hernias occur in the first 5 years after the original surgery.**
- **Midline vertical incisions are most often affected, and poor surgical technique, wound infection, obesity and chest infection are important predisposing factors.**
- **The diffuse bulge in the wound is best seen when the patient coughs or raises the head and shoulders from a pillow, thereby contracting the abdominal muscles.**
- **Strangulation is rare, but surgical repair is Usually advised.**
- **Again, open or laparoscopic mesh repair is possible.**



Complications of hernia

Irreducibility

- An irreducible hernia is one in which the contents cannot be manipulated back into the abdominal cavity.
- This is due to narrowing of the neck of the sac by fibrosis, distension of the contained bowel, or adhesions to the walls of the sac.

Strangulation

- The vessels supplying the bowel within a hernia may be compressed by the neck of the sac or by the constricting ring through which the hernia passes. The arterial supply is compromised and gangrene follows. Bacteria and toxins pass out through the bowel wall, causing peritonitis.
- The patient complains features of intestinal obstruction (Pain, vomiting, abdominal distension).
- The cough impulse is lost, and there is increasing evidence of circulatory collapse and sepsis

Obstruction

- An irreducible hernia may progress to intestinal obstruction. Abdominal pain, vomiting and distension signal the need for urgent operation *before* strangulation supervenes.

Management of complicated hernia

If there is no evidence of strangulation, an attempt can be made to reduce an apparently irreducible hernia by giving analgesia, putting the patient to bed with the foot of the bed elevated, and applying gentle pressure.

If the hernia does not reduce readily, emergency same day surgery is advised to avoid further complications.

Femoral hernias are the least likely hernia (of the common hernias) to be reduced in this way.

Management of complicated hernia

In infants and children, the majority of 'irreducible' inguinal hernias can, in fact, be safely reduced by a suitably trained clinician.

Small doses of intravenous opiate analgesia administered in the presence of suitably trained paediatric nursing staff can relax the child and assist with the reduction process.

Failure to reduce a hernia in this manner necessitates emergency surgery which is often more difficult than when the hernia has been reduced prior to surgery.

Management of Obstructed Hernia

- Urgent operation is indicated for all obstructed hernias, as one can never be certain that strangulation is not present.
- A CT scan is indicated in such cases, especially if an underlying malignancy is suspected, such as anaemia, significant weight loss or palpable mass away from the hernia.
- At surgery, the hernial sac is opened and the contents are inspected carefully.
- If they are viable, they can be returned to the abdominal cavity and the hernia repaired. If there is doubt about the viability of a loop of bowel or omentum, the devitalized tissue must be resected before proceeding to repair.
- The use of mesh like any prosthesis, can become infected. Often this requires removal of the mesh, with increased risk of recurrence of the hernia.

Thank



You